CSE 4241: Biomedical Engineering

Assignment 2: Analysis and Comparison of Research Papers in Bioinformatics

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Sources

This assignment analyzes and compares the following two research papers:

[1] M. Mongia, M. Guler, and H. Mohimani, "An Interpretable Machine Learning Approach to Identify Mechanism of Action of Antibiotics," *Scientific Reports*, vol. 12, no. 1, p. 10342, Jun. 2022, doi: 10.1038/s41598-022-14229-3.

[2] T. Warnow, Y. Tabatabaee, and S. N. Evans, "Advances in Estimating Level-1 Phylogenetic Networks from Unrooted SNPs," *Journal of Computational Biology*, vol. 32, no. 1, pp. 3–27, Jan. 2025, doi: 10.1089/cmb.2024.0710.

Contents

1 Part 1: Understanding the Papers

1.1 Paper 1 — An Interpretable Machine Learning Approach to Identify Mechanism of Action of Antibiotics

1.1.1 Problem and Its Importance

Antibiotic resistance is one of the most urgent public health crises of our time. It currently kills around 700,000 people per year worldwide and, if left unchecked, is projected to surpass cancer as a leading cause of death by 2050, claiming over ten million lives annually. The core challenge is not just finding molecules that kill bacteria, but identifying *why* they work — specifically, their mechanism of action (MOA). Existing deep learning models, like the graph neural network used by Stokes et al., could screen millions of molecules and flag bioactive ones, but they were completely uninterpretable: they could not tell researchers which part of the molecule was responsible for killing bacteria, nor could they distinguish between molecules that use already-known mechanisms versus genuinely novel ones.

This matters enormously because drug repurposing — reusing already-approved drugs against resistant bacteria — is economically attractive since those molecules have already passed safety trials. But to prioritize which repurposed molecules are truly worth pursuing, researchers need to know if they offer a new mode of attack against bacteria. Without that knowledge, laboratories waste enormous resources on pre-clinical trials for molecules that turn out to work in the exact same way as existing antibiotics.

1.1.2 Main Idea and Method

The paper introduces **InterPred**, a two-component interpretable framework. The first component predicts bioactivity; the second, called **MOACluster**, groups molecules by their mechanism of action. Rather than feeding raw molecular graphs into a black-box neural network, InterPred first extracts all unique simple ring structures and functional groups from each molecule and encodes them as binary feature vectors — a 1 if the molecule contains that substructure, a 0 if it does not. These biologically meaningful features are then fed into either a **logistic regression with L1 (lasso) regularization** or an **extra trees ensemble classifier**, both trained with a balanced scoring objective to handle the severe class imbalance (roughly 95% of molecules in the training set had no antibacterial activity).

The L1 regularization naturally drives most coefficients to zero, meaning only a small number of ring/functional group features drive the final prediction — and those features are directly readable from the model's coefficients. For MOA clustering, the top-k non-zero coefficient features reduce the feature space, and then molecules with identical reduced feature vectors are placed in the same cluster. Crucially, InterPred achieves an AUC of 0.87 on the

Stokes et al. *E. coli* test set, compared to 0.88 for the state-of-the-art deep neural network — essentially matching accuracy while remaining fully interpretable.

1.1.3 Assumptions

InterPred makes several important assumptions. First, it assumes that bioactivity is primarily determined by the *presence or absence of simple rings and functional groups* — that the structural units captured by this binary encoding are sufficient to distinguish active from inactive molecules. Second, it assumes that molecules with the same bioactive moiety share the same or similar mechanism of action. This is what allows MOACluster to use chemical substructures as proxies for MOA — for instance, beta-lactam rings are assumed to consistently indicate cell wall synthesis inhibition. Third, it assumes that the training data is representative enough that correlations between moieties and MOAs discovered during training will generalize to unseen molecules. Finally, the logistic regression model assumes a **linear relationship** between the binary feature representation and the log-odds of bioactivity, which is a simplification given the complex nature of drug-target interactions.

1.1.4 Key Limitations

The most significant limitation is that InterPred uses a **simplified molecular representation**. By reducing a complex three-dimensional molecular structure to a flat binary vector of ring and functional group presence or absence, it ignores conformational information, atom-level context, stereochemistry, and the spatial arrangement of substructures — all of which can dramatically affect how a molecule interacts with a biological target. Second, the method cannot assign a mechanism of action to a molecule with a truly novel bioactive moiety. If the substructure responsible for a molecule's activity has never appeared in any known MOA cluster, InterPred has no reference point. Third, the dataset is relatively small (a few thousand molecules), and the method may not scale straightforwardly to chemically diverse molecular libraries without retraining. Fourth, the clustering step assigns molecules with identical reduced feature vectors to the same MOA cluster, but two molecules can share a moiety yet have different MOAs due to other structural elements — the paper itself notes exceptions where the same moiety mapped to multiple mechanisms.

1.2 Paper 2 — Advances in Estimating Level-1 Phylogenetic Networks from Unrooted SNPs

1.2.1 Problem and Its Importance

Standard phylogenetic trees model evolution as a branching process with no mixing between lineages. But in reality, organisms undergo **horizontal gene transfer, hybridization, and**

recombination — events that create reticulate evolutionary histories. Phylogenetic networks are graphical models that can capture these events, but estimating them is notoriously difficult. Level-1 phylogenetic networks (or "galled trees"), where no two cycles share any nodes, are the simplest non-trivial extension of trees that can represent such events.

The specific problem this paper addresses is: *how do you reconstruct a level-1 phylogenetic network when your data is SNPs (single-nucleotide polymorphisms) and you do not know the ancestral state of each SNP?* Not knowing the ancestral state is the biologically realistic scenario — in most population genetics studies, researchers cannot determine which allele is ancestral and which is derived. Previous methods either required the ancestral state to be known (rooted SNPs) or lacked rigorous theoretical guarantees about whether their output was the *true* unique network. Gusfield's 2005 algorithm worked in this scenario but only proved consistency with the data, leaving open the possibility that multiple different networks could explain the same SNPs.

1.2.2 Main Idea and Method

The paper makes two major theoretical contributions and introduces a new algorithm. The first contribution is proving that the **GBP algorithm** (Gambette, Berry, Paul 2012) — previously thought to be the standard tool for this problem — *always fails* when given SNPs without known ancestral state and the network has at least one cycle of length five or more. This is because the quartet trees derivable from unrooted SNPs (called $Q(N)_r$) form a strict subset of the full quartet tree set $Q(N)$ that GBP requires, and this missing information causes GBP to return a contradiction.

The second and main contribution is the **CUPNS algorithm** (Constructing Unrooted Phylogenetic Networks from SNPs), a polynomial-time quartet-based method that correctly reconstructs the *semi-directed* version of any level-1 network with all cycles of length at least five, given SNPs that cover the bipartitions of the network. The paper also proves that both CUPNS and Gusfield's 2005 algorithm are **statistically consistent** estimators of the semi-directed network under a formal stochastic model of DNA evolution — meaning as alignment length grows, the estimated network converges to the true one with probability 1.

1.2.3 Assumptions

The method rests on several strict assumptions. First, SNPs are assumed to evolve under the **infinite sites assumption**, meaning each site mutates at most once in the entire evolutionary history — no back mutations, no convergent evolution, and no homoplasy. Second, all **cycles in the network must have length at least five** — this is not just a convenience but a hard mathematical requirement; the key theorems fail for cycles of length four or fewer. Third, the algorithm assumes access to a reliable **oracle** — an external process that correctly identifies which sites in a DNA alignment are true SNPs (two-state, homoplasy-free characters). Fourth, the SNPs must **cover all bipartitions** of the network. Fifth, the network is assumed to be at the relatively simple **level-1 complexity**; level-2 and higher networks are not handled.

1.2.4 Key Limitations

The most glaring limitation is the **restriction to level-1 networks with cycles of length $\geq$ 5**. Real evolutionary histories may have overlapping recombination events (cycles sharing nodes, i.e., level-2 and above), and the authors acknowledge that level-2 networks may not even be identifiable from the data. Second, the **oracle assumption** is fundamentally idealistic. In practice, no method perfectly identifies homoplasy-free sites; real datasets have noise, sequencing errors, and parallel mutations. Third, the **CUPNS algorithm has a higher runtime** than Gusfield's algorithm — $O(mn^4)$ versus $O(mn + n^3)$ for Gusfield — due to the step of computing all quartet trees, which could become a bottleneck for large datasets. Fourth, the framework does not exploit partial ancestral state information when it is available, leaving potential accuracy gains on the table.

2 Part 2: Comparison

2.1 Which Paper is More Practical for Real-World Use?

Paper 1 (InterPred) is considerably more practical for immediate real-world application. The reason is straightforward: it solves a well-defined, economically critical problem — drug discovery — using data that already exists in large public databases such as CO-ADD and the FDA-approved drug library. Biologists and pharmaceutical chemists can run InterPred using the RDKit open-source cheminformatics package without requiring specialized phylogenetic expertise. The model outputs directly actionable insights: which chemical moiety is likely responsible for bioactivity, and which cluster of molecules might share a novel mechanism. Clinicians and researchers do not need to understand logistic regression to use the tool — the output is expressed in the language of chemistry (beta-lactam rings, quinolones, guanidines) that medicinal chemists already understand intuitively.

Paper 2, while theoretically rigorous, has a more indirect path to practical application. Its utility depends on the availability of high-quality SNP alignments, a reliable oracle for identifying homoplasy-free sites, and organisms whose evolution actually fits the level-1 structure with all cycles of length ≥ 5. These are demanding requirements that may not be met in many real-world datasets. Additionally, the method is fundamentally theoretical — the paper proves properties of algorithms but does not, for instance, benchmark CUPNS on real biological datasets compared to competing methods.

2.2 Which Paper is More Theoretically Strong?

Paper 2 is far more theoretically rigorous. It provides formal mathematical proofs for every major claim: it proves that GBP fails (Theorem 3), proves that CUPNS is correct (Theorems 4 and 5), and proves statistical consistency of both CUPNS and Gusfield's algorithm under a formal stochastic model (Theorems 6–8). The paper uses combinatorial graph theory and probabilistic arguments together, working through lemma-by-lemma constructions, sample complexity bounds, and runtime analyses. The level of mathematical formalism — involving quartet trees, bipartitions, cycle topology, and measure-theoretic convergence — places this paper firmly in the domain of theoretical computer science and combinatorics.

Paper 1, by contrast, is primarily an applied engineering paper. Its core claims are validated empirically through AUC comparisons and qualitative analysis of known MOA linkages. There is some mathematical structure in the training objective and regularization, but the paper does not prove theoretical properties like consistency or sample complexity bounds. The justification for why InterPred works is largely observational rather than derived from first principles.

2.3 Which Method is More Sensitive to Data Quality?

Both methods are sensitive to data quality, but in different ways and for different reasons, as summarized in Table 1.

Table 1: Sensitivity to Data Quality: Paper 1 vs. Paper 2

Dimension	**Paper 1 (InterPred)**	**Paper 2 (CUPNS/Gusfield)**
Type of sensitivity	Mislabeled bioactivity, class imbalance, missing functional group annotations	Violation of infinite sites assumption, incorrect oracle identification of SNPs
Consequence of poor data	Model learns wrong feature-MOA correlations; clustering fails	Theorems break down entirely; output network may be wrong or algorithm returns FAIL
Severity	Gradual, measurable degradation in AUC and cluster quality	Potentially catastrophic — consistency guarantees collapse
Recovery strategy	Balanced scoring, cross-validation, L1 regularization	Partial tolerance to oracle errors shown under some conditions

Paper 2 is ultimately *more* sensitive to data quality in a qualitative sense. The mathematical guarantees of CUPNS and Gusfield's algorithm are contingent on strict conditions: the oracle must be correct, SNPs must be truly homoplasy-free, and the network must satisfy the cycle length constraint. If any of these conditions are violated, the proofs become inapplicable. InterPred, on the other hand, degrades gracefully — poor data will reduce AUC and cluster purity, but the model will still produce a prediction, and techniques like cross-validation help detect overfitting.

3 Part 3: Brainstorming & Critical Thinking

3.1 Combining Ideas from Both Papers

Here is a concrete and genuinely useful hybrid idea: **phylogenetically-guided antibiotic mechanism discovery.**

Consider the problem of tracking how antibiotic resistance evolves across bacterial strains. InterPred identifies bioactive moieties and clusters molecules by MOA, but it treats each molecule in isolation — there is no notion of evolutionary relationships between the bacteria being targeted. Meanwhile, Paper 2's framework reconstructs how bacterial populations are evolutionarily related using SNPs, capturing hybridization and lateral gene transfer events.

These two ideas can be combined as follows. First, use CUPNS or Gusfield's algorithm to construct a level-1 phylogenetic network for a set of bacterial strains, using their SNP data. This network reveals which strains are most closely related and which may have exchanged genetic material (reticulation events). Next, overlay the bioactivity data from InterPred onto the leaves (strains) of this network — specifically, which chemical moieties are effective against which strains. By analyzing patterns on the network, one could ask: do strains connected by a reticulation event show correlated changes in susceptibility to specific moieties? If a horizontal gene transfer event (reticulation) coincides with resistance emerging against beta-lactam rings, that tells you the gene transfer likely carried a resistance mechanism for cell wall synthesis inhibitors.

This combination would allow researchers to prioritize molecules with MOAs that target the *ancestral* mechanisms shared across many branches of the network, making resistance less likely to spread via reticulation — a fundamentally evolutionary-informed drug discovery strategy.

3.2 Effects of Incomplete or Noisy Data

3.2.1 Paper 1 (InterPred)

Problems: If the molecular bioactivity labels are noisy (e.g., activity is reported as inactive due to a failed assay rather than true inactivity), the logistic regression model will learn incorrect feature-MOA correlations. Since InterPred uses only 32 functional groups and ring structures, missing or incorrectly extracted substructures will silently corrupt the binary feature vectors. The balanced scoring objective helps with class imbalance but cannot correct for systematic labeling errors.

Suggested Improvement: Incorporate **label noise-robust learning** — for instance, using a noise transition matrix in the loss function (based on estimated error rates in the bioactivity assay) to discount uncertain labels. Additionally, using multiple independent feature extraction tools (not just RDKit) and taking an ensemble of feature vectors would reduce the impact of individual extraction errors.

3.2.2 Paper 2 (CUPNS/Gusfield)

Problems: If the oracle incorrectly labels a homoplastic site as an SNP, or misses a true SNP, the bipartition coverage guarantee breaks down. The infinite sites assumption is violated by parallel mutations, which are common in rapidly evolving organisms like viruses. If the input SNPs do not cover all bipartitions of the network, CUPNS will return FAIL or reconstruct an incorrect network.
Suggested Improvement: Replace the theoretical oracle with a **probabilistic model selection criterion** — for example, applying a maximum parsimony or model-based test to score each site's likelihood of being homoplasy-free, and then using a weighted CUPNS variant that assigns higher trust to sites with strong evidence of satisfying the infinite sites assumption. This would make the algorithm more tolerant of oracle imperfection, which the paper identifies as an open area of future work.

3.3 A New Idea or Application Inspired by These Papers

Proposed Application: EvoResist — Interpretable Evolutionary Drug Resistance Mapping for ESKAPE Pathogens
The ESKAPE pathogens (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* species) are the primary drivers of hospital-acquired antibiotic-resistant infections worldwide. Both papers directly engage with several of these organisms — Paper 1 tests InterPred on bioactivity data against *S. aureus*, *P. aeruginosa*, *A. baumannii*, *K. pneumoniae*, *E. coli*, and two fungal pathogens.
The proposed pipeline, **EvoResist**, operates in the following steps:

1. Collect SNP alignments from multiple clinical isolates of an ESKAPE pathogen over time (longitudinal hospital surveillance data).

2. Apply CUPNS to reconstruct the level-1 phylogenetic network of these isolates, revealing which strains exchanged genetic material with others (reticulation events likely representing plasmid-mediated resistance transfer).

3. For each leaf (isolate), collect its bioactivity profile against a panel of chemical moieties and apply InterPred's clustering to group isolates by their resistance mechanism profile.
4. Map the MOA cluster assignments onto the phylogenetic network nodes.
5. Apply a graph-based detection algorithm on the network to identify **reticulation nodes that coincide with MOA cluster transitions** — these are the evolutionary "hotspots" where new resistance mechanisms enter the population via horizontal gene transfer.
6. Output a ranked list of chemical moieties that remain effective against strains across *all* branches of the network — the most evolutionarily robust drug candidates.

This pipeline would help hospitals anticipate which resistance mechanisms are likely to spread next and which drug combinations would be hardest to develop resistance against — a genuinely novel integration of interpretable machine learning with phylogenetic network theory.

4 Part 4: Application

Option A: How Can Paper 1 Help in Discovering New Antibiotics?
Drug discovery pipelines have traditionally relied on two separate and expensive steps: first, high-throughput screening to find bioactive molecules, and then downstream mechanistic studies to determine how those molecules work. The bottleneck is the second step — determining mechanism of action is costly, slow, and requires specialized biochemical expertise. InterPred directly attacks this bottleneck. Below is a concrete description of how it could accelerate antibiotic discovery in practice.

4.0.1 Large-Scale Virtual Screening with Interpretability

Apply InterPred to chemical libraries containing millions of commercially available compounds (such as ZINC, PubChem, or ChemSpider). Unlike the Stokes et al. D-MPNN, InterPred not only flags which molecules are likely bioactive but simultaneously identifies *which functional group or ring structure* is responsible for that bioactivity. This narrows down a list of hundreds of thousands of predicted actives into a manageable set grouped by their likely mechanism, making the output tractable for experimental follow-up.

4.0.2 Prioritizing Novel Mechanisms of Action

This is InterPred's most unique contribution to drug discovery. If a molecule is predicted to be bioactive but contains a moiety that has never appeared in any known MOA cluster, it is flagged as a priority candidate for novel mechanism discovery. This is the exact kind of molecule that drug-resistant bacteria are least likely to have evolved defenses against, because the mechanism is new to them. In the CO-ADD dataset analysis, the paper already demonstrated that guanidine, nitro groups, and hydrazide derivatives show consistent patterns of bioactivity across multiple ESKAPE pathogens — and these are moieties that appear in molecules not originally designed as antibiotics.

4.0.3 Drug Repurposing at Scale

Because InterPred was also trained on an FDA-approved drug library, it can identify existing approved drugs that share bioactive moieties with known antibiotics but were originally designed for entirely different diseases. This dramatically reduces the time to clinical trials — a drug that has already passed safety testing only needs to be re-evaluated for antibiotic efficacy and dosing. Halicin, famously identified by Stokes et al. as a diabetes drug with antibiotic activity, is a perfect example of this strategy. InterPred would go further by explaining *why* Halicin works and suggesting structurally similar molecules that might work even better or through different mechanisms.

4.0.4 Designing Combination Therapies

MOACluster groups molecules by mechanism, which means it can also identify pairs of molecules in *different* clusters — molecules that attack bacteria in two completely different ways. Combination therapies using two drugs with distinct MOAs are much harder for bacteria to develop resistance to, since simultaneous mutations in two independent resistance pathways are exponentially less likely than single mutations. InterPred's clustering output directly enables the computational design of such combination strategies, which is one of the most promising directions in modern antibiotic development.

5 Part 5: Reflection

5.1 Which Paper Was More Difficult to Understand?

Paper 2 (the phylogenetic networks paper) was significantly more difficult to understand, and being honest about why is more useful than pretending both were equally accessible. The difficulty comes from multiple layers of abstraction stacked on top of each other. The paper requires fluency in graph theory — particularly directed acyclic graphs, bipartitions, quartet trees, and the notion of semi-directed networks — before any of the actual algorithms can be understood. Then it requires familiarity with population genetics and the infinite sites assumption. Then it requires reading formal mathematical proofs, many of which involve subtle case analyses (e.g., one-sided vs. two-sided cycles, cycles where the root is or is not the root of the whole network).

What made it particularly challenging is that the paper builds on a chain of prior work — Gusfield 2005, Gambette et al. 2012, Baños 2019 — and frequently references results from those papers without fully re-deriving them. A reader unfamiliar with that literature has to simultaneously understand the current contribution and reconstruct what earlier papers established. In contrast, Paper 1 is written in a style that is more accessible: the problem is stated clearly in terms of antibiotic resistance (a well-known public health issue), the algorithm is described with diagrams showing each step, and the results are reported with familiar metrics like ROC curves and AUC values.

5.2 Which Paper is More Useful for Learning?

Both papers are useful in different ways, but **Paper 1 is more immediately applicable** to the kind of work involved in machine learning and medical AI. The way InterPred combines feature engineering (extracting biologically meaningful substructures), classical machine learning (logistic regression with L1 regularization), and interpretability (reading MOA from model coefficients) is a paradigm directly transferable to other domains — including multimodal fusion models for medical image classification. The paper's treatment of class imbalance using balanced scoring is directly relevant to medical datasets where disease cases are rare relative to healthy cases.

Paper 2, however, taught something more fundamental: the importance of proving that an algorithm is *correct* and *consistent* before ever running it on real data. The fact that the GBP algorithm — a published, polynomial-time method — turns out to *always fail* on the biologically realistic input is a striking reminder that empirical success on benchmark datasets is not a substitute for theoretical guarantees. This lesson applies broadly to deep learning research: one should think more carefully about what models are theoretically guaranteed to do, not just what they empirically appear to do.

5.3 Suggested Improvement for One Paper

For **Paper 2**, I would suggest incorporating a **simulation-based validation section**. The paper is rich in theoretical guarantees, but it never tests CUPNS or Gusfield-Estimate-Unrooted on simulated or real biological data. The authors prove statistical consistency — that accuracy improves as the number of SNPs grows — but we do not know how many SNPs are practically needed for, say, 20 taxa, or how quickly accuracy degrades when the oracle incorrectly classifies 5% or 10% of sites. A simulation study with a range of network sizes, cycle lengths, alignment lengths, and oracle error rates would make the paper dramatically more useful to practitioners who want to apply these methods to real sequencing data. It would also reveal whether the theoretical $O(mn + n^3)$ runtime advantage of Gusfield's algorithm over CUPNS translates to practical differences at realistic dataset sizes. Without this empirical grounding, the paper remains an important theoretical contribution but leaves a significant gap between the proofs and actual biological application.